

1. Introduction and Motivation

Project Overview

Perfect-like constructions (PLCs) show systematic cross-linguistic patterns shaped by morphology-semantics mapping, language-specific TAM landscapes, and potentially other interacting factors. This project seeks to develop a feature-based typological model using parallel corpora and computational pipelines to evaluate these patterns across languages. Classifier outputs are interpreted through inferential statistics (χ^2 , ϕ), weighted error diagnostics, and supplemented with feature-based semantic mapping (PCA), producing results that are both computationally rigorous and linguistically interpretable. While the methodology has a computational backbone (data science, NLP, machine learning), the ultimate goal is linguistic: refining our understanding of PLCs, highlighting underdescribed variation, and connecting computational tools back to linguistic theory.

Aims & Research Questions

The project pursues four mutually reinforcing aims:

1. **Feature schema development.** Construct and iteratively refine a cross-linguistic PLC feature inventory, grounded in TAM literature and corpus evidence, with diagnostics for interpreting feature values. Corpora such as the Bible and TED Talks will provide initial data for testing and refinement, and scalability testing will incorporate data from the DoReCo corpora.
2. **Cross-disciplinary integration.** Demonstrate how computational pipelines can sharpen linguistic questions while linguistic theory can enrich computational models, producing a roadmap for sustained collaboration between linguistics and NLP.
3. **Ontological innovation.** Pilot an ontological feature schema in machine learning, encoding distinctions such as experiential versus resultative into a structured, machine-readable ontology. This allows model performance to be interpreted in typologically and semantically meaningful terms, and provides a prototype for applying ontology-driven schemas to other domains (e.g. clause type contrasts, TAM ambiguity, low-resource languages).
4. **Linguistic insight.** Ensure that the computational framework remains linguistically informed, while also generating new evidence to inform debates in TAM typology, language description, and theory.

These aims are guided by two core research questions:

1. To what extent can a feature-based semantic ontology, developed from aligned corpus data, model cross-linguistic variation in PLCs?
2. Does the application of such a schema reveal typologically consistent contrasts (e.g. experientiality, evidentiality, so-called “current relevance”) across languages that differ morphosyntactically and genealogically, providing a scalable framework for typological comparison, semantic annotation, and temporally complex meanings?

Concomitant operational questions include:

- Is the schema internally coherent and usable across languages?
- Can the features be clearly defined and consistently annotated?

- Are the features salient enough to be recognized across typologically diverse languages?
- How do individual languages encode these features, and what feature “profiles” emerge across the dataset?
- Are there recurring patterns or shared strategies across unrelated languages?

Together, these questions position the project at the intersection of typology, description, and computation, asking not only whether PLCs form a coherent cross-linguistic class (cf. Dahl & Wälchli 2016), but also whether a feature-based approach can make that coherence (or lack thereof) explicit, interpretable, and empirically testable.

Background and Literature Review

Perfect-Like Constructions: A Cross-Linguistic Puzzle

Perfect-like constructions (PLCs) sit at the intersection of tense, aspect, modality (TAM), and information structure, yet resist tidy definition even within individual languages, let alone across them. While many languages possess TAM categories that have been compared to the English present perfect (e.g. French *passé composé*, German *Perfekt*, Turkish evidentials, Mandarin *guò*), the semantic contributions of these forms vary widely, often defying one-to-one translation and challenging conventional comparative frameworks (Dahl 2000; Dahl & Wälchli 2016; Bertrand et al. 2022).

Despite their apparent shared characteristics, such as links to past events, non-narrative temporal contours, or present relevance (“current relevance”, “discourse salience”, etc.), PLCs have long been recognized as “slippery” (Plungian 2011; Schaden 2009), a quality reflected in the ongoing theoretical debate over their core semantics. This variation, both within and across languages, is precisely what makes PLCs such a compelling object of typological and computational inquiry. Their semantic subtlety and morphosyntactic variability pose significant challenges for formal theory, and offer a valuable testing ground for empirically grounded, feature-driven models.

Theoretical Foundations: Core Semantic Readings of the Perfect

PLCs have typically been associated with a small set of recurring interpretations: experiential, resultative, universal, and (less consistently) recent past readings. These readings, while not always discrete in practice, have provided a useful descriptive framework for comparing TAM constructions across languages (Comrie 1976; McCoard 1976; Klein 1992; Mittwoch 2008).

Among these, experientiality, i.e. the encoding of an event as having occurred at least once prior to reference time, has been widely cited, often via existential quantification (e.g. *I have eaten durian [before, three times]*). Resultatives, by contrast, imply a change-of-state event whose resultant state holds at reference time (e.g. *the window has broken, the butter has melted*). Universals extend the predicate across a continuous time span up to the reference point (e.g. *I have lived here for five years, she has always had brown hair*). While these categories reflect theoretical abstraction, they remain widely used in typological and computational work due to their practical utility in annotation, corpus analysis, and NLP applications (de Swart 2016; Zhao 2022a, 2022b).

The current project takes these readings not as absolute categories, but as semantic features, i.e. not mutually exclusive, but potentially co-occurring, that provide a broad and flexible categorization of interpretations. This reflects a move away from discrete categories toward feature-based modeling approaches (Bertrand et al. 2022; Haspelmath 2010a, 2010b).

Cross-Linguistic Approaches: From Slippery Categories to Predictive Features

The typological complexity of PLCs has been extensively documented in survey work and large-scale comparative projects. Seminal work by Dahl (2000) and Dahl & Velupillai (2013) established broad patterns in TAM systems across languages, while more recent studies (e.g. Bertrand et al. 2022; Zhao 2022a, 2022b; Bowler & Ozkan 2017) have emphasized the gradient, overlapping nature of PLC-like constructions across genetically and areally diverse systems.

A key insight from this literature is that cross-linguistic variation is often more a matter of degree than kind, which is to say that many languages encode “perfect-like” meanings without grammaticalizing a perfect per se. The overlap or convergence between perfects and iamitives (Dahl & Wälchli 2016), as well as the often fuzzy boundary between resultative and evidential readings (Plungian 2011), reinforce the need for modeling strategies that can tolerate polysemy and capture nuance without overcommitting to universal categories.

This has led to a recent wave of work attempting to extract interpretable, cross-linguistically predictive features of TAM categories, a trend this project builds upon. Of particular relevance is the work of van der Klis et al. (2017, 2021), LeBruyn et al. (2019), and Zhao (2022a, 2022b), who propose annotation schemas and classification models that incorporate semantic, pragmatic, and discourse-related factors. However, these models often rely on fixed inventories of TAM values or restrict themselves to genealogically related languages. This project diverges by embracing a scalable, feature-oriented approach informed by cross-linguistic corpus data and grounded in a lightweight semantic ontology.

Methodological Orientation: Toward Semantics-Lite Ontologies

The goal of this project is not so much to contribute to the *formal semantics* of the perfect per se, though it is certainly indebted to it (Portner 2003, 2011; Kratzer 2004; Iatridou et al. 2001). Instead, it aims to develop a usable, interpretable feature set that can track variation in PLCs across typologically distinct systems, and that can be productively deployed in both research and applied NLP contexts, facilitating dialog between these domains.

This approach aligns with recent calls in linguistic typology and computational semantics for hybrid models, ones that acknowledge both the utility of comparative concepts and the importance of language-specific, construction-based patterns (Haspelmath 2010a, 2010b, 2016). By grounding feature definitions in observable diagnostics, and by testing them against real corpus data, this work aims to strike a balance between cross-linguistic abstraction and empirical attunement. In doing so, it hopes to extend the toolkit available for modeling temporally complex meaning in natural language, while also providing a more interpretable alternative to opaque, large-model-derived representations (cf. Chen et al. 2021).

IV. Pilot Experiment

This section presents the pilot experiment which has been completed, and which informs and shapes the following work of the project.

Motivation and Goals

Perfect-like constructions pose challenges for both linguistic typology and computational modeling. Their semantic and morphosyntactic complexity makes them difficult to capture through traditional cross-linguistic comparison,

while their variability also poses difficulties for current NLP approaches. The pilot study therefore tests whether a feature-based annotation schema, operationalized in classifier models, can capture this complexity in a systematic and interpretable way.

The pilot aimed to assess whether the schema could be consistently applied across languages, and whether classifiers could detect meaningful patterns in feature–error relationships. This was tested by training Random Forest and MLPClassifier models on feature-annotated PLC data, and evaluating model performance through both statistical and linguistic error analysis.

As the first implementation of the schema, the pilot provides a foundation for scaling to more languages and refining the schema iteratively. The insights gained here inform methodological frameworks for subsequent experiments, including empirically grounded adjustments to feature design and annotation strategies.

This work builds on research in typological approaches to TAM (e.g. Bertrand et al. 2021; Zhao 2022a & 2022b; LeBruyn, van der Klis, & de Swart 2019; van der Klis, LeBruyn, & de Swart 2021) and computational studies of error analysis (Belnikov & Glass 2019; Geva, Goldberg, & Berant 2019; Ribeiro, Singh, & Guestrin 2016). It extends both by applying a novel feature-based methodology to cross-linguistic corpus data. The following sections detail the methodology and results of the pilot study.

Methodology

The pilot experiment uses a small aligned corpus of New Testament Bible (NTB) texts in English, German, and Mandarin. A total of 300 tokens were extracted, each representing a distinct instance of a perfect-like construction (PLC). Tokens were identified in the English NTB using a Python script that locates all instances of auxiliary *have* in co-occurrence with a verbal noun (as defined by spaCy; almost exclusively past passive participles). The script compiles these instances by verse number, which then serves as an index for aligning corresponding verses across languages. For example, verse 10010101 in English, Mandarin, and German is grouped together, enabling side-by-side comparison of PLCs in context.

Because this extraction method is anchored in English (*have* + participle), it risks bias toward forms overtly marked in English. To mitigate this, subsequent experiments will employ *reverse lookup* techniques. In this approach, English PLCs are first aligned with non-English texts, and the constructions that most consistently co-occur with English perfects (e.g. the German *Perfekt*, Mandarin *le*, Georgian *turmeobiti*) are identified. These constructions can then be used as search terms in the reverse direction, retrieving all their instances in non-English texts and examining the corresponding English alignments. Conceptually straightforward, reverse lookup is technically nontrivial: alignment noise, multifunctional markers, and the need for statistical thresholds (e.g. association strength, coverage) require careful filtering for reliable results.

Once extracted and aligned, the data are manually annotated. In this project, “annotation” refers to two processes: (i) coding each PLC instance for semantic features, and (ii) identifying which construction is used in each language (e.g., the German *Plusquamperfekt* or the Mandarin particle *guò*). Feature coding uses the English text as an “anchor,” assuming that aligned constructions encode equivalent features via translational correspondence. A crucial distinction exists between inherent semantic features (properties encoded by the construction itself) and contextual/construal features (interpretations that arise in use). While the current pilot does not attempt a systematic separation of these layers, ambiguous cases where context appears to override inherent encoding are

flagged in the metadata for reference in error analysis. A fuller implementation of this distinction is an important direction for future work.

Finally, the pilot annotation was conducted by a single L1 English speaker, which may not fully capture nuances encoded in the non-English texts. This limitation will be addressed in subsequent studies by recruiting L1 research assistants to conduct the non-English construction identification and annotation, working with a clear annotation manual and checks via inter-annotator agreement (IAA).

Features

The features used in the pilot experiment are all drawn directly from the formal semantic literature on perfects, and represent some of the most commonly cited meaning components associated with PLCs cross-linguistically. Note that one of the most commonly cited of these meaning components is what is often referred to as “current relevance”, or sometimes “discourse salience”, i.e. the implication that the event or state encoded by the predicate which is framed as a PLC carries some specific relevance for the present discourse. This feature is generally associated specifically with present perfects, and is arguably a highly salient component of their meaning in some languages (including English); however, as discussed by Grønn and von Stechow (2021), there has as yet been no widely agreed upon formal semantic definition of this feature, and there is open debate among researchers as to whether current relevance is even part of the truth conditional semantics of the perfect. For these reasons, this feature was not included in the pilot experiment, though its incorporation in subsequent experiments — in some form or other — remains a goal.

Each feature used in the pilot experiment is presented here along with the operationalized definition used to evaluate its presence or absence in each individual token. Further discussion of the performance and impact of performance on feature scheme refinement is discussed in depth in the Results and Discussion section below.

Experiential Reading¹

One of the “core” readings of the perfect is the experiential, which indicates that subject of the predicate has the experience of performing or undergoing the event at least once within a time span that is anterior to the reference time (or Topic Time) established by the perfect construction (Klein 1992). This includes expressions like *she has read this book before*, and *Padma has visited Bangkok many times*. The operationalized definition of this feature which is used in the pilot study is as follows:

A construction expresses an experiential reading if it conveys that at least one instance of the eventuality described by the predicate has occurred prior to the reference time, without necessarily implying an ongoing result state.

Because both the experiential and resultative readings fall under the umbrella term “existential”, as both readings involve existential quantification, clear diagnostics are needed to distinguish them. The first diagnostic makes use of frequency or experience markers, such as *never*, *often*, and *ever* (Mittwoch 2008, Ritz 2012): *I have often visited that park*, *she has never read that book*, and *they have been to Paris three times* all exemplify this diagnostic. Similarly, experientials are incompatible with modifiers that target result states, as exemplified in (1) below.

¹ In cases where constructions were genuinely ambiguous between experiential and resultative readings (approximately 5% of 300 pilot tokens), both features were marked as positive. These instances were tagged in the metadata to allow for downstream filtering or exclusion during analysis.

- (1) a. I have been to Tokyo, and therefore am familiar with its metro system. (experiential only)
 b. I have eaten the rice, so now it's gone. (resultative only)

Kiparsky (2002) and Ritz (2012) further observe that experiential readings are generally infelicitous with *wh*-questions — unless the question targets the event itself, rather than a resulting state. For example, in (2a) *where* targets the event of hiding, whereas in (2b) it targets the present location resulting from the hiding event, rather than the event itself.

- (2) a. Where have you found my watch? (experiential only)
 b. Where have you hidden my watch? (slightly ambiguous but strongly favors resultative)

In the pilot, constructions were annotated as experiential if they reported one or more past event instances without entailing a present result state, and were compatible with quantificational adverbs or narrative summary contexts.

Resultative Reading²

Another “core” reading of the perfect is the resultative, which indicates that the occurrence of a change-of-state event anterior to the reference time instantiates a result state that holds at that reference time (Partee 1973; Alexiadou et al. 2003). This includes expressions like *he has broken his arm [and it is now broken]* and *they have eaten all the cake [so now there is none left]*. The operationalized definition of this feature which is used in the pilot study is as follows:

A construction expresses a resultative reading if it encodes or implies a state that results from a change-of-state event, and that that state holds up to, and potentially includes, the reference time (e.g., the speech time in the case of present perfects).

Several of the diagnostics used above to disambiguate experientials from resultatives can be used in reverse to disambiguate resultatives from experientials. For example, resultatives are compatible with result state modifiers, as shown in (3) below (Partee 1973; Kratzer 2000; Alexiadou et al. 2003).

- (3) a. I have broken the vase, and it's in pieces (resultative only)
 b. He has read that book [before], #and now it's in the state of having been read (result state nonsensical)

Likewise, resultatives are not felicitous with frequency or experience markers, leading to the distinction seen in (4).

- (4) a. I have sprained my ankle twice before. (does not entail current result state; experiential only)
 b. *I have ever broken my leg. (ungrammatical; experiential only)
 c. I have broken my leg [and it is now broken]. (resultative)

An additional diagnostic is the incompatibility of resultatives with *for* adverbials, which distinguishes resultatives from universals, which are felicitous with *for* adverbials, as seen in (5).

- (5) a. I have lived here for five years. (universal and/or stative)
 b. *I have broken my leg for five years. (ungrammatical)

² See footnote __2__.

In the pilot, constructions were annotated as resultative if they entailed a state persisting from a change-of-state event and were compatible with modifiers indicating present relevance or result.

Universal Reading

The last of the “core” perfect readings included in this feature set³ is the universal, which unlike the experiential and resultative use of existential quantification instead operates on universal quantification over a time span (hence the name), which is often but not exclusively identified overtly in the context. This includes expressions like *I have always lived here*, *she has had blue hair since last month*, and *they have been reading*. The operationalized definition of this feature which is used in the pilot study is as follows:

A construction expresses a universal reading if it conveys that a state or event holds continuously throughout a contextually defined time span, typically up to (but not necessarily including) the reference time.

Perhaps the most commonly used diagnostic for identifying universal perfects makes use of their compatibility with *for* adverbials (Comrie 1976; Iatridou et al. 2003), as exemplified in (6).

- (6) a. She has studied Latin for five years. (*for* adverbial felicitous → universal)
b. #He has broken his leg for five years. (non-durable event incompatible with *for* adverbial)

Another prominent characteristic of universal perfects is their durative predicate class restrictions: universals are only felicitous with states and activities, i.e. atelic/durative predicates (Dowty 1979; Comrie 1976). This is exemplified in (7).

- (7) a. I have always been tall. (state → universal available)
b. She has been swimming for an hour. (activity → universal available)
c. #They have painted the house for seven years. (universal available only if interpreted as habitual activity)
d. #He has arrived for three days. (achievement → universal unavailable)

Thirdly, (English) universals allow for paraphrasing of *since* adverbials with *all along*⁴ (Comrie 1976; Iatridou et al. 2003), as shown in (8).

- (8) a. She has known Anto *since childhood*. → She has known Anto *all along*. (universal available)
b. She has visited Anto *since college*. → She has visited Anto *all along*. (iterative, not universal)

It has also been noted (Schaden 2009; Dahl & Velupillai 2013; Plungian & van der Auwera 2006) that many languages do not encode universal readings via PLC at all, encoding universal-type readings present tenses, generally accompanied by progressive or habitual aspect.

³ The fourth is the “recent past” reading, which is generally treated as a pragmatic nuance of the experiential and resultative readings. While occasionally marked morphologically in some languages, in most systems it is more often inferred from context and adverbial modification, such as *just* (Comrie 1976; Kiparsky 2002; McCawley 1971; Iatridou et al. 2003; Dahl 1985)..

⁴ The specific temporal anchor is lost, but the continuity or persistence of the event (“temporal contour”) remains available through contextual inference.

In the pilot annotation, universal readings were identified where the predicate described a state or activity holding continuously over a contextually bounded interval (Comrie 1976; Iatridou et al. 2003). These readings were restricted to durative predicates and were often difficult to distinguish from stative presents in translation.

Compatibility with DTAM

This feature refers to the ability of a present-situated PLC to co-occur felicitously with (past referencing) definite temporal adverbial modification (DTAM), such as *yesterday*, *in 1999*, or *last week*; i.e. time expressions which refer to a specific, bounded point or interval in the past. The operationalized definition of this feature which is used in the pilot study is as follows:

A token is annotated as positive for DTAM if it permits felicitous co-occurrence with such adverbials.

This feature is of particular interest in the study of PLCs due to the well-documented “Present Perfect Puzzle” (Klein 1992; Pancheva 2004; Pancheva & von Stechow 2004; many others), i.e. the differing language-specific compatibility of present-situated PLCs with DTAM. This is exemplified in (9)⁵.

- (9) a. **ENG** You heard it yesterday (*have heard).
b. **SPA** Ya lo oísteis ayer (*habéis oído).
c. **DEU** Ihr hat es gestern gehört.
d. **HBS** To ste juče čuli.

Non-English examples will be glossed in the submission draft.

In the pilot dataset, present perfects in English were uniformly annotated as negative for DTAM due to the nature of the extraction process. This bias will be addressed in future iterations via reverse-lookup methodologies, as described in the introduction to this section.

Evidentiality

Evidentiality refers to linguistic strategies for conveying the source of information or knowledge conveyed in a given utterance. In some languages grammatical systems, evidentiality is overtly encoded, such as via a dedicated morpheme or truth conditional semantics of certain constructions more generally, while in others (such as English), it is expressed by means of circumlocutions such as *apparently*, *I hear (that...)*, or *they told me (that...)*. The operationalized definition of this feature which is used in the pilot study is as follows:

A construction expresses evidentiality if it overtly marks the source of information, either through grammatical markers or lexical/discourse framing (e.g. *apparently*, *I heard*, *they say*).

The examples in (10) demonstrate evidentiality in English, which does not overtly mark evidentiality.

- (10) English
a. He has eaten all the cake.
b. Apparently he has eaten all the cake.
c. I heard he has eaten all the cake.

⁵ Umberto Eco, ParaSol Corpus, via Grønn & von Stechow (2021).

In (10), though English does not grammatically encode evidentiality, each expression conveys a differing attitude of the speaker toward the information conveyed (including, pragmatically, degree of confidence in the information). In (10a) the information is conveyed as epistemically neutral; no source of information is specified, and the assumption is that the information is factual. In (10b) while no specific source of information is expressed, the use of *apparently* provides the speaker with a certain amount of confidence “wobble room”; the information is plausible, even probable, but the speaker is not committing to it entirely, i.e. epistemic hedging. In (10c) the source of information is not expressly named, the speaker is expressly placing confidence in the information on an outside source.

Next consider the examples in (11), from Georgian, a language which does grammatically encode evidentiality.

(11) Georgian

- a. mteli namtskhvari sheuch'amia. *he has (apparently) eaten all the cake*
- b. mteli namtskhvari shech'muli akvs. *he has eaten all the cake*

Non-English examples will be glossed in the submission draft.

Both examples in (11) can be faithfully translated in English as *he has eaten all the cake*, though each conveys a different evidential meaning. (11a) uses the Synthetic Perfect (*turmeobiti* I), which intrinsically conveys an evidential stance such that the information expressed is gleaned either through hearsay or deduced from visual evidence. This expression is only felicitous in contexts where one or both of these factors is present. By contrast, (11b) uses the periphrastic perfect, which is epistemically neutral and felicitous under any evidential context.

Various languages that are slated for possible inclusion in subsequent experiments (e.g. Georgian⁶, Turkish⁷, and Bulgarian⁸) are noted for conveying evidentially in their PLCs specifically. In fact, as noted in the typological literature (e.g. Aikhenvald 2004), evidentiality is frequently grammaticalized in many languages, and in some cases is particularly associated with PLCs, and thus this feature is included in the schema used in this project.

As evidentiality is not grammatically encoded in the English TAM system, the handful of instances of this feature in the pilot experimental dataset all either explicitly encode source of knowledge (such as “he saw that [predicate]”) or otherwise make source of knowledge explicit in the discourse context.

TAM Layering

Tense-Aspect-Modality Layering (TAML) refers to constructions in which multiple TAM operators co-occur within a single clause, often combining to yield complex interpretive effects. In the context of PLCs, this commonly includes combinations such as perfect + progressive (e.g. *has been running*), perfect + past (e.g. *had eaten*), or perfect + modal (e.g. *might/should/could have arrived*). The feature schema used in this project treats such combinations as a distinct structural property with potential semantic effects. The operationalized definition of this feature which is used in the pilot study is as follows:

A construction incorporates TAM layering if it combines two or more tense-aspect-modality components (e.g., perfect + past, perfect + progressive, perfect + modal).

⁶ Widely cited, incl. Makharoblidze (2012), Kurtsikidze (2006), Bolkvadze & Kiziria (2023)

⁷ Şener (2011), de Gül (2008), Rett (2013)

⁸ Rett & Hyams (2013)

While not a distinct semantic category in and of itself, such as resultativity or evidentiality, TAML often interacts with semantic categories typologically associated with PLCs. In some languages, TAML is grammaticalized and semantically restricted, as in Japanese *-te ita* (Nishiyama & Keonig 2010; Clarke 2013) and Georgian pluperfect/*turmeobiti* II (Makharoblidze 2012; Kurtsikidze 2006; Bolkvadze & Kiziria 2023). Typologically, TAML is constrained but non-trivially variable, and PLCs are frequently at the center of such variation (Cinque 1999; Bybee et al. 1994; Bohnemeyer 2014; Iatridou et al. 2003).

This feature also reflects a broader orientation within this project, which treats PLCs less as isolated semantic readings (e.g., experiential, resultative) and more as structurally complex tense constructions. This perspective informs both the typological framing and methodological design of the study.

In the pilot experiment this feature was a major source of classifier prediction errors; this will be discussed in more depth in the Results and Discussion section below.

Negation/Polarity Effects

While not a canonical diagnostic of PLCs, there is cross-linguistic evidence that polarity can interact with TAM expression, especially in systems with multiple PLCs (e.g. Georgian), such that negative polarity may favor one form over the other due to such factors as scope, inference blocking, and evidential contrasts (Iatridou et al. 2003; Michaelis 1994; Portner 2003). The operationalized definition of this feature which is used in the pilot study is as follows:

A token is marked as expressing negation if the construction appears under the scope of a negator, lexically, morphologically, or syntactically. This surface-level annotation is intended to assess whether constructional choice varies as a function of polarity.

The initial motivation to include this feature in project derives from the frequently cited (Makharoblidze 2012; Hewitt 1996; Aronson 1990; Harris 1982; Kurtsikidze 2006; and others) property of the Georgian Synthetic Perfect (Series III *turmeobiti* I) such that it is preferred over both the Aorist and Periphrastic Perfect under scope of negation. Georgian is considered a typologically interesting case study for this project as a rare language which bears multiple PLCs in its TAM system. As such the feature was incorporated on an exploratory basis.

In annotating polarity, I adopted a surface-oriented approach, marking constructions as negative if they appear under the scope of an overt negator. However, it is worth noting that many non-veridical contexts, such as interrogatives, conditionals, and counterfactuals, also emerged in the pilot dataset and may warrant further investigation. These environments are traditionally associated with polarity sensitivity (Giannakidou 1998; Israel 2011), and it is plausible that they interact with TAM selection in subtle but systematic ways across languages.

While these contexts are not annotated separately in the current pilot, they are noted in the metadata and may serve as the basis for expanded polarity-related analysis in future iterations of the study.

Honorary Mention: Current Relevance

A final note is warranted on the often-cited but methodologically elusive notion of “current relevance.” Especially in the context of the English present perfect, current relevance has long been proposed as a defining semantic contribution linking past events to present discourse salience (e.g. McCawley 1971; Comrie 1976; Michaelis 1994). However, despite its ubiquity in the descriptive literature, there is no widely accepted diagnostic for identifying

current relevance in context, nor consensus on whether it is part of the truth-conditional content of the construction at all (see Portner 2003; Kiparsky 2002; Schaden 2009; Grønn & von Stechow 2021).

Because this study relies on discrete, scorable features that can be operationalized for cross-linguistic comparison, current relevance is not included in the present feature schema. Nonetheless, its persistence in the literature, and its relevance (current or otherwise) to how speakers and researchers conceptualize PLCs, warrants its mention here as a direction for further qualitative inquiry. Feedback and suggestions on this topic are very welcome.

Construction Identification

One of the greatest — but also most intriguing, at least for this morphology-obsessed researcher — challenges in the annotation process was identifying which construction was used in each language, how to group (or split) those constructions into categories, as well as whether an aligned token in the non-English data was “equivalent”, meaning that it encoded the core semantics⁹ that are associated with PLCs. In this section I describe the constructions observed in each of the languages of the study, how they were annotated, and the motivations behind their respective categorizations.

English

As the anchor language used in this experiment, every token which was extracted from the corpus for evaluation was expressed in English as a perfect, i.e. auxiliary *have* + participle. These perfects were then divided by the specific TAM profile in which it was encoded, yielding six constructions: [present] perfect, past perfect, *would* perfect, *could* perfect, *should* perfect, and *might* perfect. By far the most numerous were the perfect and past perfect categories; these were identified by the morphological tense expressed on the auxiliary, *have/has* for perfect and *had* for past perfects. No examples of progressive or other overt aspectual components were evident in the data (aside from the aspectual property conveyed by the perfect itself), though should such examples arise in subsequent experimental data, they will be annotated as such (creating a new construction designation for each, such as “present perfect progressive”), on the premise that it is easier to reduce distinctions later as needed than it is to re-annotate an entire dataset for distinctions that were not distinguished in the first pass.

Building on the same logic, each of the modal perfects was annotated distinctly, yielding a four-way distinction between *would*, *could*, *should*, and *might* perfects. The underlying motivation for distinguishing between these constructions is that while each can be described as simply a “modal perfect”, it is possible that differences in cross-linguistic TAM systems may give rise to unforeseen interactions between modality and PLC selection. Based on the results of the pilot study, reflecting, as it does, a relatively small sample size of 300 total tokens (150 used for training and 150 for testing), it is unclear whether such fine-grained distinctions are in fact warranted in the grand scheme. As such, they will continue to be distinguished from one another until such time as experimental results either more robustly motivate their inclusion or indicate that they introduce more “noise” into the data than truly informative contrasts.

German

The German annotations recognized eight distinct constructions, each being associated with a specific morphological profile¹⁰: Präsens, Präteritum, Perfekt, Plusquamperfekt, Konjunktiv I Perfekt, Konjunktiv II Präteritum, Konjunktiv II Perfekt, and Konjunktiv II Plusquamperfekt. Constructions in the data were identified

⁹ By “core semantics” I refer to being compatible with at least one of the three canonical interpretations of PLCs: the experiential, the resultative, and the universal.

¹⁰ All constructions are Indikativ (indicative) unless otherwise noted.

using a combination of grammars and digital parsing tools (including Google Translate and the [Reverso.net](https://www.reverso.net) German conjugator). Because each construction is associated with a distinct semantic profile, they were kept distinct, though as with the English modal perfects, ongoing experimental results may indicate that some of these distinctions can (or should) be collapsed.

As the data used in the pilot experiment (i.e. New Testament Bibles) reflects a distinctive and at times archaic register, one “construction” did appear which is no longer used in Modern German, specifically a bare participle. There are a tiny handful¹¹ of examples of what I term “null aux” PLCs in German, such as the one shown in (12).

(12) nachdem ich allem von vorn an genau **nachgegangen** *after I have traced everything from the beginning*

Non-English examples will be glossed in the submission draft.

In this example, from verse number 42001003, the participle *nachgegangen* occurs with no auxiliary either locally or in the wider clause. Based on input from several L1 German speakers, this construction is considered archaic, and no longer used in the present day. In such cases, the construction is coded as a Perfekt based on the observation that each corresponded to an English [present] perfect, though the peculiarity was recorded in the metadata.

Mandarin

Categorizing the Mandarin data proved to be somewhat challenging, as the TAM system it utilizes is quite distinct from what is found in the (vast majority of the) Indo-European languages on which much of the work on PLCs has been conducted, including those of strongly isolating languages such as English. Mandarin, rather than overtly marking tense in the conventional sense of definite time reference (i.e. past, present, future, non-past, etc.), it instead uses a variety of strategies for conveying the temporal properties of predicates.

One strategy that occurred quite frequently in the experimental dataset is what is annotated “unmarked”, i.e. a bare verb with no temporal situation or contour conveyed. Approximately 15% of tokens in the data utilized this strategy in correspondence with the English perfect. As the term “unmarked” carries specific connotations in the linguistics literature (an oversight on the part of the experimenter!), henceforth the constructions annotated as “unmarked” will instead be referred to as “nullmarked”, reserving “unmarked” for the more standard use as something akin to “default”. In the case of Mandarin, as well as other languages with similar TAM systems that utilize null TAM marking, a compelling puzzle arises: what is the “unmarked” interpretation of a predicate, with respect to the temporal properties? This facet of the linguistic landscape will be considered carefully in ongoing study.

Another common strategy is the use of grammatical particles which are often referred to as “aspectual” in nature; the two most commonly associated with perfect-like meanings are *le* (often described as “perfective” or “completive”) and *guò* (sometimes referred to as “experiential”), both of which occur post-verbally. In addition to these, the light verb *yǒu* ‘have’ and its negative polarity counterpart *méiyǒu* are seen with some frequency in correspondence with the English perfect.

¹¹ In fact, in the finalized dataset that was submitted to the classifier, there was only one example, though perhaps two to three were found in an earlier annotation phase which was not ultimately included in the pilot dataset.

In addition to these, the construction *suǒ...de*, where the two elements more or less bookend the predicate, also appeared somewhat frequently in the dataset. This construction is used as something like a passive/relative clause marker, as seen in (13).

- (13) Mandarin *suǒ...de* examples
a. *suǒ chī de* *that which is eaten*
b. *suǒ zhīdào de* *that which is known*

Non-English examples will be glossed in the submission draft.

While not strictly speaking a TAM marker, the construction does appear somewhat frequently in correspondence with the English perfect.

In addition to these, Mandarin makes extensive use of adverbials to convey the temporal situation and contours of predicates. Many of these convey imitative (CITE) meanings, similar to English *already*, but come in a wide variety of subtle distinctions, including *yǐjīng* (neutral, “default” *already*-like meaning), *zǎoyǐ* (more emphatic and formal, with a remote past connotation), and *qǐlái* (less *already*, more directional/resultative with an inchoative dimension as well).

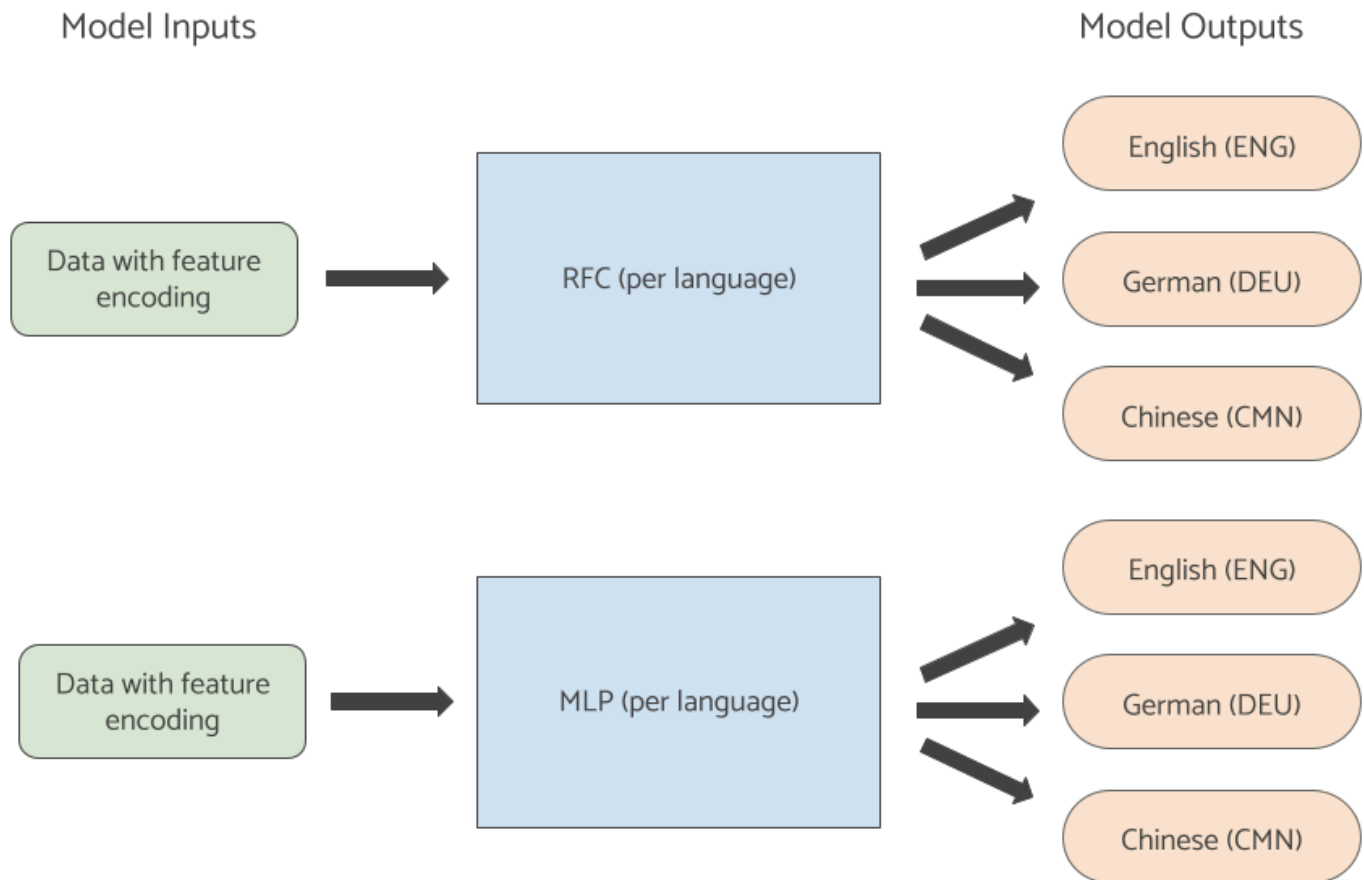
Further complicating the playing field, many of these components occur in tandem, such as *méiyǒu...guò, yǒu...le*, and [adverb] *le*. In the pilot annotation phase, each of these constructions was annotated distinctly, again on the intuition that it is easier to collapse than to expand distinctions later, yielding a total of 18 constructions in the Mandarin data. This diversity of in many cases relatively low-frequency constructions no doubt added to the classifiers’ struggles in accurately predicting construction occurrence. Moving forward, this list of constructions will almost certainly be flattened at least slightly, though analysis of their distribution and the profiles of the errors associated with each construction is a work in progress.

It should be noted that roughly 8% of the Mandarin tokens in the experimental data were marked as “N/A”, as like with the “nullmarked” tokens they lacked overt TAM marking. These two groups were distinguished (somewhat informally, though as rigorously as possible given the limitations noted in the introduction to this methodology section) by whether some element of temporal/aspectual specification or contrast could be recovered from context. This approach is obviously suboptimal, and will be addressed going forward by involving L1 Mandarin speakers in evaluating the data.

Last but not least, the Mandarin TAM system employs a strategy for aspectual specification whereby a “lexical aspect marker” (my term) is used to convey a specific aspectual meaning. To clarify, these are not markers of lexical aspect, but rather lexical markers of aspect. In essence, how this system works is that a semantically contentful, usually monosyllabic root verb is joined with something like a like verb, often with directional/motion qualities, which thus “lexically” engenders a specific aspectual nuance. In the pilot experiment, all such examples, both those otherwise overtly marked and those not were omitted from consideration based on the relatively limited expertise of the annotator with respect to effectively evaluating the nuances conveyed. Again, this will be addressed going forward through incorporation of the expertise of L1 speakers.

Running the Experiments

Once all of the data had been annotated, they were fed into two classifier models for experimentation: Random Forests (“RFC”) and MLPClassifier (“MLP”). Random Forests operates on a decision tree model, while MLPClassifier (Multi-Layer Perceptron) is a neural network; using the two classifiers in tandem allows for examining the performance of the feature set using two distinct model types, thus providing greater context for evaluation of their salience and framing. In other words, a separate classifier was trained for each language, so that performance could be evaluated independently across language–classifier pairs.



Workflow for classifier training and prediction. Feature-encoded data were used to train a separate Random Forest (RFC) and Multi-Layer Perceptron (MLP) classifier for each language (ENG, DEU, CMN), with outputs reported independently by language/classifier pair.

Both experiments used the same strategy for loading in the data: the 300 tokens were split into a 150-token training set and a 150-token test set. Although this pilot split was random rather than stratified, subsequent experiments will explicitly stratify by feature to ensure proportional distribution. The classifiers then used the 150-token training set to “learn” the features and correlation patterns with occurrence of constructions, and based on that training attempted to predict which constructions would be used in test set tokens. The results of the predictions were then exported for use in error analysis. Note that both the German and Mandarin datasets included tokens coded as “not applicable”, indicating that the token for that language was not usable for some reason; in subsequent experiments this binary applicable/ not-applicable distinction will be further decomposed

into at least two subtypes of N/A: tokens in which a given verb/predicate is simply absent in that language's text ("true" N/A), and tokens in which it is unclear whether the corresponding token "truly" encodes a PLC.

Results and Discussion

Overview

Before diving into language-specific results, one generalization can be made regarding both the models' performance and also the methodology that produced the model inputs. The overall accuracy rates by language were, perhaps unsurprisingly, quite different. The English datasets both contained all 150 tokens, and performed with accuracy rates¹² of 92.7% for RFC and 91.3% for MLP. German accuracy was lower than that of English, with accuracy rates of 75.8% for RFC and 76.4%. Both models achieved significantly lower accuracy on the Mandarin data, with accuracy rates of 54.8% for RFC and 40.4% for MLP. There are a variety of factors which may be affecting this discrepancy between languages; here I highlight two specifically: anchor language effects and feature landscape effects.

Anchor Language Effects

Because English was used as the anchor language for this experiment, naturally every English token in the initial 300-token set was included, unlike with the German and Mandarin which in some cases were deemed non-equivalent. Even more significant, however, is the fact that the feature values for each token were assigned based on the English anchor text, meaning that the mapping between feature profile and construction in English is generally reliable; certainly there are ambiguous and other edge cases, but by and large the diagnostics used to evaluate feature values for each token are fairly well fleshed out and presumed to more or less accurately represent the features encoded by each token's PLC.

By contrast, the German and to an even greater extent the Mandarin data presumably encode feature nuances that are not present in the English data, and perhaps even unrepresented in the feature set itself. While English and German exhibit significant differences in their respective TAM systems, they do share common linguistic ancestry (both being Low German languages) and have continued to interact consistently since their divergence from a common root language approximately 1500 years ago. As such, there are certain commonalities in the logic on which each language's TAM system operates: both languages overtly mark tense; they make similar use modal verbs (e.g. *can*, *should*, *may*, etc.), and many their modals are cognates; both languages encode future "tense" using a modal verb (English uses *will* and German uses *werden*); both languages form PLCs using an auxiliary (*have* in English and *have* or *be* in German, depending on argument structure) and a participle; while the English subjunctive mood has atrophied considerably, both languages display sensitivity to mood contrasts; as well as various other similarities. The tension between similarities and differences in the English and German TAM systems may explain German's accuracy rate being higher than that of Mandarin but not as high as that of English.

Mandarin, on the other hand, is wholly unrelated to English (or German), and its TAM system operates on a logic which is entirely distinct from what is seen in Indo-European languages generally, to say nothing of Low Germanic languages specifically. As described above, English and German TAM marking systems (i.e. morphologies) bear certain fundamental similarities, such as overt marking of tense. Mandarin, on the other hand, does not overtly mark tense at all, favoring instead a system by which tense (i.e. deictic temporal "location") is inferred from context. Additionally, while Mandarin makes extensive use of aspectual particles and adverbials to convey temporal

¹² Here "accuracy rates" refers to F1-scores.

contours, bare (null-marked) verbs are licit in many contexts, while conversely English and German verbs require some temporal specification in their expression¹³.

Subsequent experiments will explore ways to reduce anchor language effects, such as incorporating reserve lookup across all languages, L1 speaker input, and feature refinement. These measures cannot fully eliminate anchor effects, and indeed may risk redistributing bias rather than removing it, but they do represent important steps toward testing how robust the methodology can be across typologically diverse languages.

Feature Landscape Effects

As mentioned in the previous section, with respect to the Mandarin TAM system, it generally goes without saying that each language's TAM (in fact, entire grammatical system) operates on its own logic, including what features it chooses to contrast or mark overtly versus those it doesn't¹⁴, the number of TAM constructions in the system's inventory, and more. Because the bulk of the research conducted on PLCs to date has focused on Indo-European, and especially Germanic languages (English alone represents a significant focus of PLC research historically), the feature set that was developed for the pilot experiment, reflecting research landscape, tends to better reflect the types of features that those languages "care about". For example, if the English perfect is considered a "complex tense"¹⁵, i.e. in the Reichenbachian (1947) sense of a grammatical form which combines tense and aspect, or which divides its event and reference times, to what extent can we truly apply the logic developed for English to a language which, like Mandarin, has been called "tenseless" (Lin 2006 and 2010, Collart 2021)?

Tackling this quandary is one of the central aims of this project, by way of developing a feature set which can be used effectively on cross-linguistic and typologically varied languages.

In the next section I present the results by language along with some discussion, followed by some global/cross-linguistic performance discussion.

Results and Interpretations by Language

This section presents the results of each language-classifier experiment. The primary metric is overall performance accuracy (how well did the classifier do globally, what is its success rate), which I will examine using several key methods, including specific error types, feature-error type associations, and several statistical evaluations (including χ^2 scores).

Performance Metrics

The overall performance accuracy is generated as a report within the classifier experiment script itself, returning for each language-classifier pair four values: **precision** tells us, of the tokens the model predicted as this construction, how many were correct; **recall** refers to the ratio of the number of times a construction occurred in the dataset to the number of times the model correctly predicted that construction; the **f1-score** is the harmonic mean of precision and recall used to balance between the two; **support** is the (whole) number of occurrences of

¹³ English does allow ostensibly bare verbs in the case of habituais (*we eat breakfast [every day]*) and statives (*I like salads*), though a case can be made that in these instances tense is expressed through null morphology.

¹⁴ Of course, this reflects an etic perspective (researcher-imposed categories) rather than an emic one (language-internal categories).

¹⁵ This is my own term, but I'm quite fond of it. I am unsure to what extent it is used elsewhere in the literature; I have seen it, but not a lot.

the construction in the test dataset. These values are generated (and reported here) for each construction found in the experimental dataset as well as for each language-classifier pair's overall performance.

“Specific error types” refers to a specific, erroneous pairing of predicted to actual construction. For example, the error type denoted here as $le \rightarrow \text{nullmarked}$ is an error type in which the model predicted le when the actual construction was nullmarked. While a variety of distinct error types (and patterns of error types) were observed in the experimental results, here focus will be placed on those error types deemed most informative to the ongoing project.

An important lens for uncovering potential patterns in the models' errors involves examining associations between features and error types. While the primary metric for performance evaluation is the overall accuracy of the models' performance, knowing that, for example, RFC mispredicted le for actual nullmarked tokens eight times in the experiment tells one part of the story, which is then given more context and perspective through examination of the feature values associated with that error. One example of this is the Mandarin-RFC errors $le \rightarrow \text{suo...de}$ and $le \rightarrow \text{nullmarked}$, in which nearly all $le \rightarrow \text{suo...de}$ errors were accomplishments and nearly all $le \rightarrow \text{nullmarked}$ errors were achievements. This pattern suggests that the accomplishment/achievement distinction in Mandarin plays a potentially significant role in the model's ability to “learn” and make predictions about those constructions.

The metrics presented so far and then subjected to statistical analysis in order to assess whether certain associations between, say, features and error types, are statistically significant to the models' performance. Three statistical metrics were employed in the performance analysis. The first is a χ^2 test of features-to-errors, which is used to determine whether a model's accuracy/error rate with respect to specific features is statistically significant in the form of p -values, where a p -value of <0.05 is considered significant. Importantly, these tests do not compare one construction label to another; rather, they test the association between a feature's presence/absence and the likelihood of error overall. These values were calculated along a 2×2 axis comparing positive and negative occurrence of features and errors; i.e. $f+ e+$ represents the number of tokens that bear a feature and were predicted erroneously, while $f- e+$ represents the number of tokens that do not bear a feature and were predicted erroneously. This analysis incorporated the Yates correction, which is designed to correct for smaller sample sizes to avoid overstating significance values.

Another significance evaluation method applied to the experimental results is in the form of ϕ -values, which complement p -values such that while p -values assess whether an association (between feature and error rates) is statistically reliable, ϕ -values show the magnitude or “strength” of that association. This measure is particularly useful in the context of smaller sample sizes, such as that used for the pilot experiment, as p -values are sensitive to sample size, while ϕ -values are independent of sample size. Unlike WS (below), ϕ -values are non-directional: they capture only the strength of the association, not whether the feature made prediction easier or harder. ϕ -values are calculated using the square root of the value of the χ^2 -score divided by the total sample size for that table. To sum up, these two values let us see not only whether a feature-error link is significant (p) but also how strong that link is (ϕ).

Finally, a custom measure is used called a “weighted score” (WS), which measures the impact of a feature relative to a hypothetical baseline performance (the term “baseline” is further explained below). The WS scales with how frequent a given feature is, and is sensitive both to error bias and to how frequently the feature occurs (i.e. the practical importance of a feature in the dataset). The WS can be positive or negative, with a lower score indicating that the model performed better on a feature and a higher score indicating weaker performance on a feature. With respect to the WS, “baseline” refers to the expected error rate under uniform distribution, calculated as total errors

divided by total tokens, and a value of 0 indicates performance indistinguishable from baseline (no evidence that the feature makes prediction easier or harder). This provides a neutral reference point against which feature-level performance can be evaluated.

Thus, WS is a directional measure, in contrast to ϕ -values, which are not. The WS is complemented by a “range-normalized score” (RNS) which places the WS in the context of the ranges of WS’s for that language-classifier pair, yielding a 0–1 value, where 0 indicates the highest rate of success by associated feature and 1 the lowest within that language-classifier pair. This allows for a limited but informative degree of comparison of feature-error performance across language-classifier pairs. See Appendix A for an explanation of how these values are calculated.

All feature-level metrics (e.g. FER¹⁶, WS) are calculated per feature; language-level results are reported separately as overall accuracy (not an aggregation of feature scores). Thus, feature metrics provide diagnostics within a language, while accuracy situates overall performance at the language level. Table 1 provides a comparison of per-language and per-feature metrics.

Feature	FER	WS	RNS
Resultative	37.3%	4.34	1.00
Experiential	70.0%	-5.68	0.00
Accomplishment	42.3%	-1.32	0.44
Achievement	44.7%	-0.22	0.55
Overall accuracy (Mandarin RFC)		45.2%	

Example Table Showing Language+Classifier-Specific Feature Performance

These metrics are summarized in (14) for ease of reference.

(14) Statistical metrics

- The p -value indicates the statistical reliability of an association between feature(s) and error(s).
- The ϕ -value indicates the strength of association between feature(s) and error(s), independent of sample size; the scale used presumes 0-0.29 = weak association, 0.3-0.49 = medium association, 0.5+ = strong association.
- The weighted score (WS) represents the relative difficulty posed by a given feature to the models’ predictive efforts.
- The range-normalized score (RNS) makes results cross-comparable across languages/models with very different numeric ranges; the higher a value the worse a model performed and vice versa.
- Taking the The ϕ -value and WS together allows us to evaluate both how strongly a feature is associated with predictive errors (ϕ) and the degree to which it matters in practice (WS).

¹⁶ “Feature-specific error rate”, the number of times an error is tagged for a given feature.

Mandarin Results

The RFC experiment achieved an overall accuracy of 55% on the Mandarin data, reflecting 63 accurate predictions out of 115 total predictions. The MLP experiment achieved an overall accuracy of 50% on the Mandarin data, reflecting 61 accurate predictions out of 121 total predictions.

Major Error Types

Predicted → Actual	Count	% of all errors
le → suo_de	7	13%
le → nullmarked	8	15%
le → yijing_le	8	15%
nullmarked → le	7	13%
nullmarked → (varies)	6	12%
Totals:	36	69%

Table NUM: Major Mandarin RFC Errors

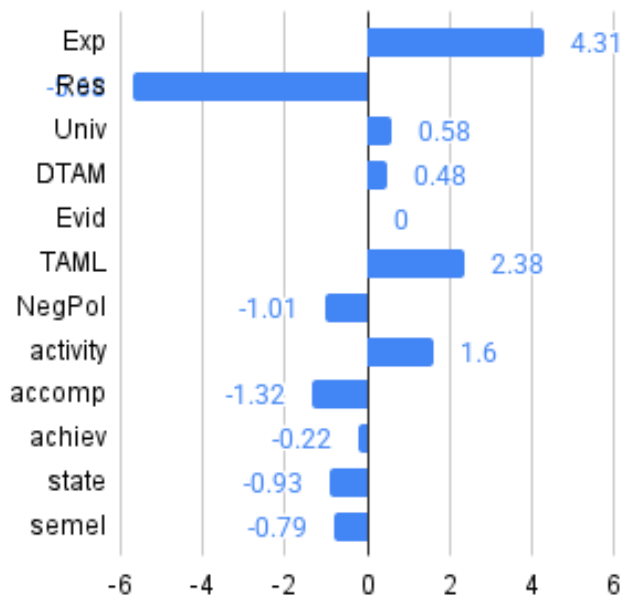
Predicted → Actual	Count	% of all errors
le → suo_de	9	15%
le → nullmarked	8	13%
le → yijing_le	8	13%
suo_de → (varies)	14	23%
nullmarked → le	5	8%
nullmarked → (varies)	5	8%
Totals:	49	82%

Table NUM: Major Mandarin MLP Errors

Both classifiers show a tendency to overgeneralize on the usage of *le*, representing 43% of all RFC errors and 41% of all MLP errors, the usage of *nullmarked*, representing 25% of all RFC errors and 17% of all MLP errors, and the usage of *suo...de*, a pattern which RFC captured at a much lower rate of three individual errors (not shown here; one was actual *guò* and two were actual *nullmarked*), suggesting a difference in how the models learn and utilize the feature schema. The 14 MLP actuals for this error type were: five instances of *le*, three of *guo*, three *nullmarked*, two *meiyou...guo*, and one of the adverbial *zaoyi*, an *already*-like iative with remote past connotations. The *nullmarked* → (varies) errors were widely distributed across construction actuals, showing no discernible pattern.

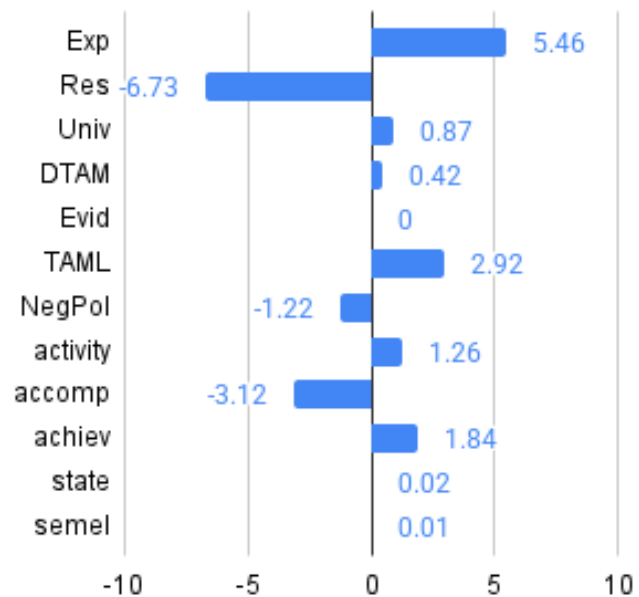
Feature Metrics

CMN RFC



CMN RFC

CMN MLP



CMN MLP

Weighted Scores (WS) By Feature

Higher scores = more difficult to predict, lower scores = easier to predict

Random Forest (RFC)

MLPClassifier (MLP)

	FER%	WS	RNS	FER%	WS	RNS
Experiential	70%	4.31	1.00	74%	5.46	1.00
Resultative	37%	-5.68	0.00	40%	-6.73	0.00
Universal	50%	0.58	0.63	58%	0.87	0.62
DTAM	100%	0.48	0.62	100%	0.42	0.59
Evidentiality	0%	0.00	0.57	0%	0.00	0.55
TAML	100%	2.38	0.81	100%	2.92	0.79
NegPol	29%	-1.01	0.47	20%	-1.22	0.45
Activity	71%	1.60	0.73	71%	1.26	0.66
Accomplishment	42%	-1.32	0.44	43%	-3.12	0.30
Achievement	45%	-0.22	0.55	54%	1.84	0.70
State	33%	-0.93	0.48	50%	0.02	0.55
Semelfactive	0%	-0.79	0.49	50%	0.01	0.55

Mandarin Feature Metrics for Random Forests and MLPClassifier

WS range for RFC: -5.68 to 4.31 (9.99). WS range for MLP: -6.73 to 5.46 (12.20)

WS color gradient runs darker for lower scores and lighter for higher scores

Both models show strongest contrasts for Experiential (hard) versus Resultative (easy), with other errors near baseline (here 0.45 for RFC and 0.50 for MLP). RFC is slightly more clustered while MLP spreads difficulty across more features. Effect sizes are weak (see next table) so interpretations remain tentative.

The tables above show overall feature performance by classifier; recall that lower WS and RNS values indicate stronger performance on that feature. Table NUM shows the statistical measures for the features deemed most significant by χ^2 analysis, informing the degree to which the feature performance rate impacted overall accuracy.

	<i>p</i> -value	ϕ -value
Resultative, RFC	0.011695	0.2546
Resultative, MLP	0.00235	0.2946
Experiential, RFC	0.027602	0.2285
Experiential, MLP	0.007608	0.2625

Statistical Measures

For both models, the Resultative and Experiential features were deemed statistically significant, though ϕ -values indicate weak association between errors and features, with the exception of the Resultative feature in the MLP experiment, which nearly breaches the threshold for medium association strength.

Summary and Discussion

A brief overview of major error types and their feature value associations:

- **Le** → **suo...de**: 13% of all RFC errors and 15% of all MLP errors; all instances of this error in both experiments were resultatives and favored accomplishments at a rate 5/7 for RFC and 6/8 for MLP, in both cases the remainder were achievements.
- **Le** → **nullmarked**: 15% of all RFC errors and 13% of all MLP errors; all instances of this error in both experiments were resultatives and favored achievements at a rate of 6/8 for both models, the remainder being accomplishments.
- **Le** → **yijing...le**: 15% of all RFC errors and 13% of all MLP errors; all but one instance of this error in both models were resultatives with one experiential exception, and were evenly split between accomplishments and achievements.
- **Nullmarked** → **(varies)**: 12% of all RFC errors and 8% of all MLP errors; feature values for both showed no salient patterning, save that tokens were evenly split between experientials and universals and included no resultatives.
- **Nullmarked** → **le**: 13% of all RFC errors and 8% of all MLP errors; feature values for both showed no salient patterning, save that tokens were evenly split between experientials and universals and included no resultatives.
- **Suo...de** → **(varies)**: 23% of all MLP errors; all instances of this error were experientials, while the remaining feature values showed no salient patterning.

With respect to error types, as noted above, the *le* → *yijing...le* errors may not in fact be not “true” errors; the models are both correctly predicting the occurrence of the particle *le*, and omitting the co-occurring adverb. This may suggest that the models are interpreting the presence of the adverb as non-consequential, though more robust data is needed to confirm this hypothesis.

The fact that *le* → *suo...de* errors are predominantly accomplishments and *le* → *nullmarked* errors are predominantly achievements is noteworthy, though a χ^2 test indicated that Aktionsart is not a significant feature in the Mandarin data, so for the time being I remain judicious with claims regarding their significance for the models.

The *nullmarked* → (varies) error is noteworthy for lacking tokens bearing the Resultative feature (they were evenly split between experientials and universals in both experiments), a feature which is borne by 83 of 115 (72%) tokens in the RFC dataset and 85 of 121 (70%) of the MLP data. It bears noting that these features are in nearly complementary distribution in the dataset, the exceptions being the 15/300 tokens that are ambiguous Experiential/Resultative or 3/115 in the Mandarin experiment specifically. This may suggest that the Mandarin TAM system makes use of a +Resultative/-Resultative distinction such that PLCs bearing the Resultative feature are treated as one class and all remaining PLCs constructions as another (nonhomogeneous) class, or at least that the models appear to interpret resultatives vs. non-resultatives in this way.

The *suo...de* → (varies) error is noteworthy for appearing only in the MLP experiment data, and also for the fact that every token in the set is an experiential such that it represents 56% of all prediction errors that bear the Experiential feature. Taken in tandem with a low *p*-value for the Experiential feature in the Mandarin MLP dataset, this suggests that experientiality is highly salient to this model, though any specific claims must remain judicious in light of this feature’s low ϕ -value.

While the *le* → *nullmarked* and *le* → *suo...de* errors suggest a correlation with Aktionsart (accomplishment and achievement specifically), as mentioned previously the Aktionsart *p*-values for both models did not indicate feature significance. It is possible that further testing involving a larger dataset and a refined feature schema may illuminate the role of Aktionsart in the Mandarin TAM system.

The *nullmarked* → *le* and *nullmarked* → (varies) errors all exhibit relatively even feature distribution, meaning that no readily discernible patterning in feature association presented itself. These errors may represent low frequency feature value profiles, though a truly rigorous analysis of this hypothesis requires more data and testing for confirmation.

From a linguistic perspective, these results highlight the limitations of the current feature schema for capturing nuanced distinctions in the interpretations of Mandarin PLCs. Much existing PLC research focuses on Indo-European (esp. Germanic) languages, leaving non-Indo-European systems underexplored by comparison. This project has the potential to expand our typological understanding of PLCs by identifying which semantic features are most relevant to TAM marking cross-linguistically. Subsequent experiments will refine the feature schema as informed by cross-linguistic insights.¹⁷.

¹⁷ Note that this observation was absolutely included in the design logic of the feature schema used in the pilot experiments, though in the interest of “casting a wide net” the feature set was kept relatively simple.

German

The RFC experiment achieved an overall accuracy of 76% on the German data, reflecting 113 accurate predictions out of 149 total predictions. The MLP experiment also achieved an overall accuracy of 76% on the German data, reflecting 113 accurate predictions out of 148 total predictions.

Major Error Types

Predicted → Actual	Count	% of all errors
pastperf → k2_pastperf	7	19%
pastperf → past	4	11%
pastperf → perf	4	11%
perf → pres	4	11%
perf → (varies)	7	19%
pres → (varies)	4	11%
Totals:	26	84%

Table NUM: Major German RFC Errors

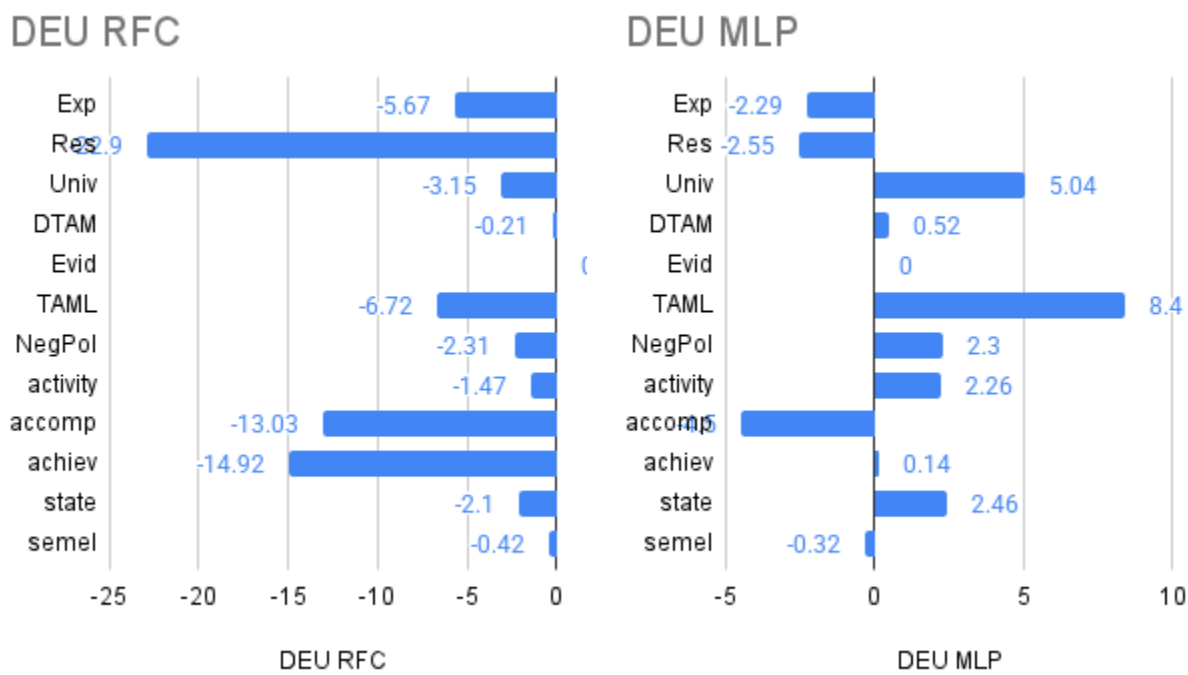
Predicted → Actual	Count	% of all errors
pastperf → k2_pastperf	6	17%
pastperf → past	4	11%
pastperf → perf	4	11%
perf → pres	4	11%
perf → (varies)	7	20%
pres → (varies)	4	11%
	29	83%

Table NUM: Major German MLP Errors

The two models performed nearly identically with respect to their major error types. One particularly noteworthy pattern is the tendency of both models to overgeneralize the use of *pastperf*, representing 50% of all RFC errors and 49% of all MLP errors¹⁸. The *perf* construction is similarly overpredicted at identical rates, with the “varies” categories having identical breakdowns: *k1_perf* (Konjunktiv I Perfekt) once, *k2_pastperf* once, *past* three times, *pastperf* twice, and *pres* four times. MLP also showed a slight tendency to overpredict *pres*, twice for *k2_pastperf*, once for *pastperf*, and once for *pres*.

¹⁸ These percentages include overpredictions of *pastperf* which are not represented in these tables.

Feature Metrics



Weighted Scores (WS) By Feature

Higher scores = more difficult to predict, lower scores = easier to predict

	Random Forest (RFC)			MLPClassifier (MLP)		
	FER%	WS	RNS	FER%	WS	RNS
Experiential	11%	-5.67	0.75	11%	-2.29	0.17
Resultative	21%	-22.90	0.00	20%	-2.55	0.15
Universal	73%	-3.15	0.86	73%	5.04	0.74
DTAM	100%	-0.21	0.99	100%	0.52	0.39
Evidentiality	0%	0.00	1.00	0%	0.00	0.35
TAML	64%	-6.72	0.71	63%	8.40	1.00
NegPol	55%	-2.31	0.90	55%	2.30	0.53
Activity	75%	-1.47	0.94	71%	2.26	0.52
Accomplishment	13%	-13.03	0.43	13%	-4.50	0.00
Achievement	24%	-14.92	0.35	24%	0.14	0.36
State	60%	-2.10	0.91	60%	2.46	0.54
Semelfactive	0%	-0.42	0.98	0%	-0.32	0.32

German Feature Metrics for Random Forests and MLPClassifier

WS range for RFC: -22.90 to 0.00 (22.90). WS range for MLP: -4.50 to 8.40 (12.90)
WS color gradient runs darker for lower scores and lighter for higher scores

Perhaps most striking in the feature metrics presented here is the difference in the ranges of the models' WS. While MLP performed similarly on German as on Mandarin, both in terms of range size and bookend values, RFC performed significantly better than baseline (0.24 for both models) on all features. Much of this variance is driven by the Resultative feature, which RFC handled very well, and to a lesser extent the Accomplishment and Achievement features.

The WS range for the German RFC experiment was -22.90 to -0.00 (22.90), indicating that all features scored below baseline, which in turn means that the model handled each feature at least somewhat better than expected. However, the distance between the easiest feature (Resultative) and the least easy (Evidentiality) is vast. These results suggest a highly uneven degree of ease: some features (Resultative) are disproportionately easy, while others are only marginally easier than baseline, pointing to strong feature-level asymmetry.

By contrast, the WS range for the German MLP experiment was -4.50 to 8.40 (12.90), indicating broader differentiation across features. In this case, some features performed worse than baseline and others better, yielding a more balanced profile than RFC. In other words, MLP highlights multiple German features as more or less error-prone, but without the extreme skew observed in RFC.

The tables above show overall feature performance by classifier; recall that lower WS and RNS values indicate stronger performance on that feature. Table NUM shows the statistical measures for the features deemed most significant by χ^2 analysis, informing the degree to which the feature performance rate impacted overall accuracy.

	<i>p</i> -value	ϕ -value
TAML RFC	0.00001	0.5051
TAML, MLP	0.00001	0.4802
Universal, RFC	0.00001	0.3943
Universal, MLP	0.00001	0.3927
Activity, RFC	0.001908	0.2894
Activity, MLP	0.009534	0.2505
State, RFC	0.014953	0.231
State, MLP	0.015683	0.2303
Accomplishment, RFC	0.017441	0.2108
Accomplishment, MLP	0.017441	0.2108
Negative Polarity, RFC	0.031157	0.2068
Negative Polarity, MLP	0.032536	0.206

Statistical Measures

For both models, the TAML and Universal features were deemed very highly significant, with TAML ϕ -values at or near the threshold for strong association strength, and Universal ϕ -values well within the medium association

strength range. While the remaining features were all deemed statistically significant, the association strength for all is weak.

Summary and Discussion

A brief overview of major error types and their feature value associations:

- **Pastperf** → **k2_pastperf**: 19% of all RFC errors and 17% of all MLP errors; all instances of this error type in both experiments bear the Resultative and TAML features, and both favor achievements at a rate of 5/7 for RFC and 5/6 for MLP.
- **Pastperf** → **past**: 11% of all RFC errors for both models; all instances of this error type in both experiments bear the Resultative and TAML features, and are all accomplishments.
- **Pastperf** → **perf**: 11% of all errors for both models; all instances of this error type in both experiments bear the Resultative and TAML features, and are all achievements.
- **Perf** → **pres**: 11% of all errors for both models; all are universals and activities.
- **Perf** → (**varies**): 19% of all RFC errors and 20% of all MLP errors; in both experiments, the values for all features appear randomly distributed.
- **Pres** → (**varies**): 11% of all errors for both models; all are universals and states.

There are several key observations which can be made on examination of these results: the two models performed nearly identically on the German test data; both models particularly struggled with universals and TAM layering; both models performed notably well with resultatives (RFC especially) though the χ^2 results were not deemed significant; regardless of statistical measurements, feature values for all but one error type (**perf** → (**varies**)), correlate very neatly with error types, which suggests that the models are able to interpret the features but struggle to associate them with actual constructions.

In the case of both models' overprediction of **pastperf**, all instances correlate directly with the TAML feature, which in turn was shown to both be statistically significant and to bear high strength of association. Taken together, this paints a picture wherein the TAML feature causes the most difficulty for both models to predict. I propose that this owes to the coarse-grainedness of the feature itself; by definition it encompasses all cases wherein a perfect bears a secondary TAM feature (in this experiment, essentially past and modal perfects). The insight that can be gleaned from this is that the TAML needs to be further decomposed into more nuanced categories such as "tense shifted" for past perfects and "modal" or even subcategories of modality for modal perfects.

The strong performance of the Resultative feature coupled with low statistical significance scores is a matter which is currently under review. Similarly, the apparent correlation patterns between Aktionsart and error type, while statistically significant, is complicated by relatively low strength of association scores. Both of these facets of the experimental results are being taken under consideration with respect to how they may inform subsequent experimental methodologies; input and feedback are very welcome!

Another pattern of note is that seen in the errors involving the Universal and Activity and State features. Both models made errors in distinguishing between the present and perfect in the presence of the Univ feature which appear to correlate with activities and states. The results show that the models erroneously predict present in the case of universal activities and perfect in the case of universal states. Based on this, it seems safe to hypothesize that in German, the perfect is more closely associated with universal activities while the present is more closely associated with universal states.

The errors of type *perf* → (varies), with their more or less random feature value distributions, most likely reflect an overgeneralization on both models' part to default to the perfect in cases where specific feature value profiles prove uninformative. This will be monitored on an ongoing basis as the feature schema, datasets, and overall experimental methodologies evolve.

English

The RFC experiment achieved an overall accuracy of 93% on the English data, reflecting 139 accurate predictions out of 150 total predictions. The MLP experiment achieved an overall accuracy of 91% on the English data, reflecting 137 accurate predictions out of 150 total predictions.

Major Error Types

Predicted → Actual	Count	% of all errors
<i>pastperf</i> → [modal] <i>perf</i>	8	73%

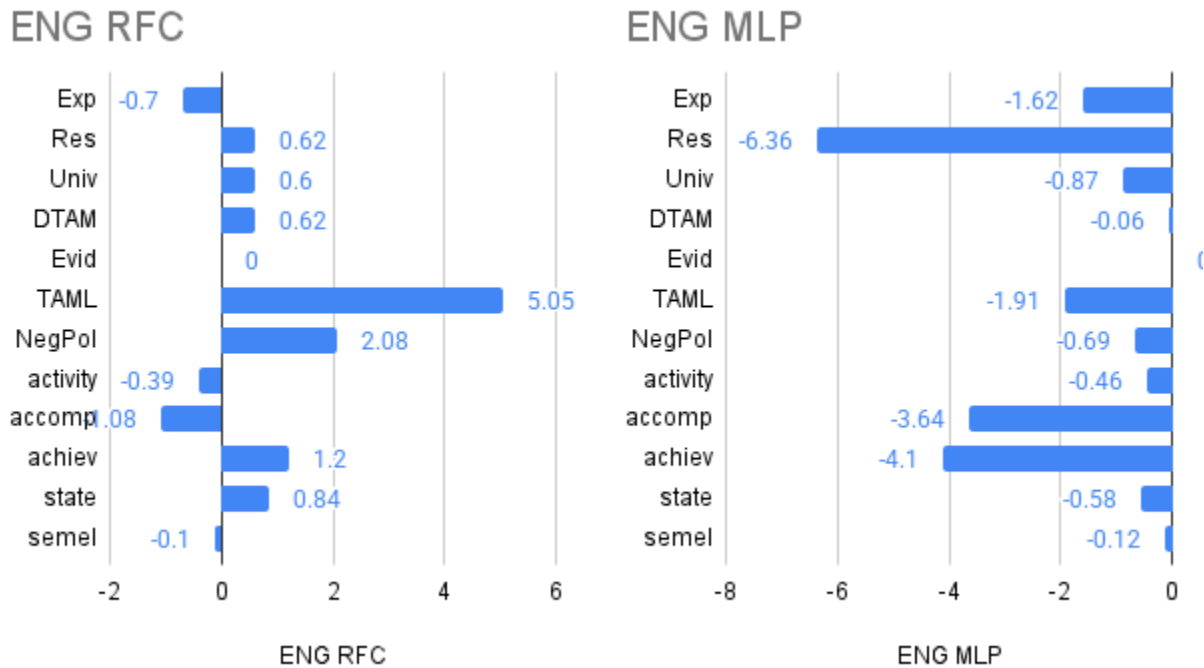
Table NUM: Major English RFC Errors

Predicted → Actual	Count	% of all errors
<i>pastperf</i> → [modal] <i>perf</i>	5	45%

Table NUM: Major English MLP Errors

Given both models' high overall accuracy rate, it is perhaps unsurprising that only one major error type emerged, though the respective occurrence rates diverged somewhat. Note that in addition to these, MLP also erroneously predicted *pastperf* for actual *perf* twice, which, if considered in tandem with the *pastperf* → [modal]*perf* error type brings the overall rate of *pastperf* overprediction roughly in line with the RFC rate.

Feature Metrics



Weighted Scores (WS) By Feature

Higher scores = more difficult to predict, lower scores = easier to predict

	Random Forest (RFC)			MLPClassifier (MLP)		
	FER%	WS	RNS	FER%	WS	RNS
Experiential	4%	-0.70	0.06	4%	-1.62	0.75
Resultative	8%	0.62	0.28	9%	-6.36	0.00
Universal	13%	0.60	0.27	13%	-0.87	0.86
DTAM	100%	0.62	0.28	100%	-0.06	0.99
Evidentiality	0%	0.00	0.18	0%	0.00	1.00
TAML	30%	5.05	1.00	33%	-1.91	0.70
NegPol	33%	2.08	0.52	33%	-0.69	0.89
Activity	0%	-0.39	0.11	13%	-0.46	0.93
Accomplishment	5%	-1.08	0.00	5%	-3.64	0.43
Achievement	10%	1.20	0.37	10%	-4.10	0.35
State	20%	0.84	0.31	20%	-0.58	0.91
Semelfactive	0%	-0.10	0.16	0%	-0.12	0.98

German Feature Metrics for Random Forests and MLPClassifier

WS range for RFC: -1.08 to 5.05 (6.13). WS range for MLP: -6.36 to 0.00 (6.36)

WS color gradient runs darker for lower scores and lighter for higher scores

Based on the data presented in the tables above, both models show feature-level differentiation, but their directional bias is inverted. On the one hand, RFC shows broad over-sensitivity, yielding worse-than-baseline performance on most English features, which suggests that the model is “noticing” features but in a way that hurts prediction. This is why so many WS values are positive, i.e. it’s disproportionately making errors on those features, as though it’s over-weighting them or latching onto misleading cues.

Conversely MLP shows broad undersensitivity, performing better than baseline across nearly all features, which suggests that the model is also differentiating features, but almost all WS are negative, meaning it is systematically making fewer errors than baseline; i.e. the model is treating those features as useful guides rather than traps, so performance improves across the board.

The tables above show overall feature performance by classifier; recall that lower WS and RNS values indicate stronger performance on that feature. Table NUM shows the statistical measures for the features deemed most significant by χ^2 analysis, informing the degree to which the feature performance rate impacted overall accuracy.

	p -value	ϕ -value
TAML RFC	0.00001	0.468
TAML, MLP	0.00001	0.4656
Negative Polarity, RFC	0.002488	0.2941
Negative Polarity, MLP	0.0085	0.2585

Statistical Measures

Only two features were deemed statistically significant by way of χ^2 test: TAML and Negative Polarity. The significance of both features is quite strong, with TAML standing out especially. The TAML ϕ -values also approach the threshold of strong association, though Negative Polarity shows rather weaker association.

Summary and Discussion

A brief overview of major error type and its feature value associations:

- **Pastperf** → [modals]: 73% of all RFC errors and 45% of all MLP errors; all instances of this error bear the TAML feature, and favor achievements at a rate of 6/8 for RFC and 4/5 for MLP.
- The remaining results for both experiments showed no discernible patterning, either by error type or feature value clustering.

Both models performed very well on the English data, perhaps unsurprising given that not only is English the anchor language of the experiment, English is also less morphologically complex than German (though more so than Mandarin), which yields a smaller set of constructions to work with. Additionally, the vast majority of the research on PLCs has at least started with if not entirely focused on English, so the feature schema in its starting state is presumably influenced by features associated with the English perfect specifically, owing to its prevalence in the PLC literature.

Aside from this, there is a clear correlation between the TAML feature prediction errors, which corroborates the insight from German that this feature warrants further decomposition. From a linguistic perspective, this is

relatively transparent: the TAML feature encompasses both tense-shifted and modal forms (*had gone*, *might/would/could/should have gone*), and the model clearly wants to assign *pastperf* to all of them; presumably it reads *pastperf* as a “default” for TAML based on frequency (or other TBD) effects.

Additionally, NegPol in both experiments shows a relatively strong p -value, and a ϕ -value in the weak to nearly-medium strength of association range, though its WS is unexceptional. My interpretation of this facet of the results is that *something* is going on with NegPol, though with a small sample size (only 12 of 150 tokens in both experiments bear this feature) it is difficult to draw serious conclusions.

General Summary and Discussion

The pilot experiment utilized 300 tokens in three languages, which, while not adequate to support major conclusions, does provide a snapshot that can inform how to proceed with the remaining stages of the project. The dataset is small, the features are preliminary and somewhat naïve, and the annotations were done by a single annotator (myself). As such, these results should be treated as indicative rather than conclusive. Nonetheless, they offer several important lessons for how to refine the project going forward.

Across languages and classifiers the dominant pattern is not diffuse error but clustering: a few feature contrasts disproportionately account for prediction failures. For Mandarin, the Experiential versus Resultative axis consistently drove classifier difficulty, while in German and English both, TAM layering (TAML) had a strong negative impact. In German, the Universal feature was strongly associated with erroneous predictions, and there are suggestive patterns involving Aktionsart, specifically, misclassifications between accomplishments and achievements in resultatives, and between activities and states in universals. In English, overlap between TAML and NegPol appears to skew the latter feature’s results, suggesting that decomposition of TAM layering and more precise coding of modality will be needed.

Error patterns further underscore the limits of the current feature schema. In Mandarin, for example, the models often predicted *le* in resultative contexts even where *suo...de* or null-marked constructions also occur frequently. This suggests that the current feature constellation is not sufficient to capture the TAM contrasts in Mandarin, where the system may be structured differently from English and German. Similarly, in German, the universals proved challenging: both models conflated perfect and present constructions, and errors showed systematic associations with Aktionsart (activities vs. states). These patterns are suggestive but not yet significant; larger and better-annotated datasets will be required to confirm them.

More broadly, these findings highlight a typological concern: can perfects/PLCs be treated as a coherent cross-linguistic class? Dahl & Wälchli (2016) argue that perfects are slippery not only cross-linguistically but also within languages, and the present results echo that difficulty. Similarly, Bertrand et al. (2022) propose decomposing PLCs into a four-way categorical distinction between experientials, resultatives, past perfectives, and “hybrids” (i.e. those which allow both experiential and resultative readings).

From one perspective this poses problems for the methodology, but from another it is extremely useful: if better-annotated data (particularly with input from L1 speakers) reinforces even some of the tentative patterns here, that would provide strong evidence for the feasibility of a feature-based typology of PLCs.

Future work will therefore focus on three fronts: (i) refining the feature schema (especially TAM layering and Aktionsart), (ii) expanding the dataset with more tokens and languages, and (iii) incorporating L1 annotators to improve the reliability of feature coding and construction equivalence judgments. These steps will help ensure that

ambiguous or edge cases are consistently treated, and will allow the project to move from pilot-scale diagnostics toward a broader typological framework.

V. Dissertation Plan

This dissertation is structured as three interlocking projects, each building on the pilot and feeding into a cumulative typological framework.

Structure of the Dissertation

Chapter 1

Methodological refinement. Extend the pilot experiment to a larger dataset, incorporating additional languages, refining the feature schema, and scaling annotation with L1 research assistants. This stage will consolidate the ontology-based methodology and establish a baseline for feature-driven error diagnostics.

Chapter 2

Model comparison. Evaluate the feature-based methodology against distributional approaches, such as vector-based embeddings (e.g. BERT, S-BERT). This comparison will assess whether ontological feature schemas provide interpretability and typological insight beyond what purely distributional models capture.

Chapter 3

Generalization and scalability. Test whether the feature-based ontology can handle novel data (previously unseen tokens/languages). This stage addresses the central question of whether the schema provides a generalizable framework for typological comparison and whether it can support extension to new domains.

Together, these projects move from refining the method (Chapter 1), to testing it against alternatives (Chapter 2), to scaling and generalizing it (Chapter 3).

Planning & Strategy

Potential Challenges & Mitigation

The most immediate challenge is annotation, which is both time-intensive and prone to subjectivity. To address this, the project will recruit and train L1 research assistants, whose contributions will improve both speed and reliability of annotation. Guidelines will be iteratively refined to ensure consistency, even in ambiguous cases, and an inter-annotator agreement (IAI) will be employed to further assure that data annotation is consistent across annotators. A broader risk is that the methodology may ultimately fail to produce strong or consistent results. Even in that scenario, however, the findings would remain informative: they would highlight the limits of current feature-based approaches and clarify which aspects of PLCs resist formalization, offering valuable insight for both typology and NLP.

Dissemination

Each project is designed to yield conference presentations and eventual publications.

- Project 1 (methodological refinement): SIGTYP and a corpus-linguistic journal.
- Project 2 (model comparison): CoNLL and Typology at the Crossroads.
- Project 3 (generalization): Possible venues include LREC (Language Resources and Evaluation Conference), COLING, or ACL workshops on low-resource languages and typology.

Timeline

The project will proceed on a compressed schedule to allow for defense by August 2026.

- Fall 2025: Finalize pilot chapter; recruit and train RAs; begin large-scale annotation.
- Winter 2026: Complete methodological refinement (Chapter 1) and submit initial conference paper. Run model comparison experiments (Chapter 2); prepare submissions to CoNLL/Typology at the Crossroads.
- Spring 2026: Conduct generalization/scalability experiments (Chapter 3); prepare submission to LREC or ACL-related venue.
- Summer 2026: Synthesize findings and finalize dissertation.
- August 2026: Dissertation defense.

Appendix A: WS and RNS Calculations

The values for WS and RNS are calculated as follows:

- Baseline error rate (BER): $\text{total errors} / \text{total tokens}$
- Feature-specific error rate (FER): $\text{errors that bear the feature} / \text{total occurrences of the feature in the dataset}$
- Feature prevalence (PREV): $\text{total occurrences of the feature in the dataset} / \text{total tokens}$
- Base weighted score (BWS): $(\text{FER} - \text{BER}) * \text{PREV}$; the subtraction compares “how hard this feature is” relative to the model’s average performance, and the multiplication by prevalence scales that difference by how frequent the feature actually is in the dataset
- Weighted score rescaled (WS): $\text{BWS} * 100$; this scales the score to be readily interpretable
- Range-normalized score (RNS): $(\text{WS} - \text{MIN}(\text{WSR for that language-model pair})) / (\text{MAX}(\text{WS for that language-model pair}) - \text{MIN}(\text{WS for that language-model pair}))$